

Everywhere-local reversibility without a global reversor

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Abstract

We construct a composition of four generalized Hénon maps over $K = \mathbb{Q}(\sqrt{7})$ that has a polynomial reversor over every completion of K , but has no polynomial reversor over K . The global exclusion allows arbitrary polynomial degrees and orders; the local witnesses happen to be affine involutions. The construction belongs to a scalar family for which we determine the entire polynomial reversing coset over every number field. On the Jung–van der Kulk tree, the degree cycle $(7, 15, 15, 7)$ forces every centralizer translation to be a power of the original map. An explicit calculation of the pointwise axis stabilizer then reduces reversibility to one eighth-power equation. The classical Wang class 16, which is an eighth power in every completion of $\mathbb{Q}(\sqrt{7})$ but not in that field, supplies the arithmetic obstruction. The contribution is its realization for the constrained pair consisting of one map and its inverse, with all alternative polynomial reversors excluded.

1 The all-place reversibility question

A polynomial automorphism can be conjugate to its inverse over every completion of a number field without being conjugate to its inverse over that field. We give an explicit example in the group of polynomial automorphisms of the affine plane. The distinction concerns existence of a polynomial reversor, not just existence of a reversor in a selected affine family.

For a field L , let $\text{Aut}_L(\mathbb{A}^2)$ denote the group of polynomial maps of the affine plane with polynomial inverses, all defined over L . Group multiplication is composition, with the rightmost map acting first. For $F \in \text{Aut}_L(\mathbb{A}^2)$, put

$$\text{Rev}_L(F) = \{R \in \text{Aut}_L(\mathbb{A}^2) : RFR^{-1} = F^{-1}\}.$$

We call F *reversible over L* when this set is nonempty. Our notation $\text{Rev}_L(F)$ includes reversors only, not the union of reversors and commuting symmetries.

Let Ω_K be the set of all finite and infinite places of a number field K . The local-global question is whether

$$[\text{Rev}_{K_v}(F) \neq \emptyset \text{ for every } v \in \Omega_K] \implies \text{Rev}_K(F) \neq \emptyset \quad (1.1)$$

for every finite nonempty composition of generalized Hénon maps

$$H_i(x, y) = (y, P_i(y) - a_i x), \quad P_i \in K[y], \quad \deg P_i \geq 2, \quad a_i \in K^\times.$$

There is no degree or order bound on a proposed reversor. No completion may be enlarged or omitted, and F is the full ordered composition, not an iterate chosen in its place.

For $c \neq 0$ and $d \geq 2$, write

$$H_{c,d}(x, y) = (y, cy^d - x). \quad (1.2)$$

Its inverse is $(x, y) \mapsto (cx^d - y, x)$. Each such map therefore satisfies the preceding Hénon hypotheses.

Theorem 1.1. *Set*

$$K = \mathbb{Q}(\sqrt{7}), \quad F = H_{4,7}H_{1/16,15}H_{16,15}H_{1/4,7}. \quad (1.3)$$

For every $v \in \Omega_K$ there is an affine involution $R_v \in \text{Aut}_{K_v}(\mathbb{A}^2)$ such that $R_v F R_v^{-1} = F^{-1}$. Nevertheless,

$$\text{Rev}_K(F) = \emptyset.$$

In particular, implication (1.1) is false even when the search for a global reversor includes all polynomial degrees and orders.

Although F has rational coefficients, the base field in Theorem 1.1 is the specified field K . Both of its infinite places are real; it has no complex places. The local witnesses and global exclusion will be proved over precisely these fields.

The main algebraic step is an exhaustive calculation for a scalar family. For a number field L and $a \in L^\times$, define

$$F_a = H_{a,7}H_{a^{-2},15}H_{a^2,15}H_{a^{-1},7}, \quad R_t(x, y) = (t^{-1}y, tx) \quad (t \neq 0). \quad (1.4)$$

Theorem 1.2. *For every number field L and every $a \in L^\times$,*

$$\text{Rev}_L(F_a) = \{F_a^j R_t : j \in \mathbb{Z}, t \in L^\times, t^8 = a^2\}. \quad (1.5)$$

Consequently, F_a is reversible over L if and only if $a^2 \in L^{\times 8}$.

The integer j in (1.5) is unrestricted. The theorem is an equality for the entire polynomial reversor set, not an affine ansatz. It will also provide the explicit local construction: whenever $t^8 = a^2$ in a characteristic-zero field, R_t reverses F_a .

Classical inputs and the distinction from conjugacy

The polynomial group structure used here is classical. The Jung–van der Kulk amalgam, its reduced words, and the associated symmetry and reversor methods are described in Gómez–Meiss [3] and Baake–Roberts [1]. We specialize that machinery to one four-factor word and prove the required full centralizer equality directly. A commuting-subgroup inclusion would not suffice. In particular, the field hypothesis on roots of unity in [1, Theorem 2 and Corollary 1] cannot be imposed during a calculation over $\mathbb{C}$.

The arithmetic obstruction is also classical. The exceptional eighth-power class over $\mathbb{Q}(\sqrt{7})$ is recorded in Song Wang’s account of the Grunwald–Wang theorem [4]; we verify every place directly below. Scalar twisting together with exhaustive centralizer descent is not a new method: the final example of Cantat–Dujardin [2, Section 2.2] already controls the field of polynomial conjugacy of two scalar-twisted Hénon maps through their full centralizer. Combining that example with the Wang class gives an analogous all-place failure for two independently specified maps. That consequence of the cited mechanisms does not identify the second map with the first map’s inverse. Here the remaining task is to realize the obstruction for the constrained pair (F, F^{-1}) and exclude every alternative reversor.

The proof first determines the entire geometric centralizer, including its translation image on the amalgam tree (Sections 2–3). Section 4 then proves Theorem 1.2 by a scalar twist and original-field descent. Section 5 proves Theorem 1.1. The two-factor control in Section 6 explains why failure of a selected affine swap cannot replace the full-group argument.

2 The full polynomial group and a primitive axis

We work over $\mathbb{C}$ in this section and the next. Let $G = \text{Aut}_{\mathbb{C}}(\mathbb{A}^2)$, let $\mathcal{A}$ be its affine subgroup, and set

$$\mathcal{E} = \{(x, y) \mapsto (\alpha x + p(y), \beta y + \eta) : \alpha\beta \neq 0, p \in \mathbb{C}[y]\}, \quad \mathcal{B} = \mathcal{A} \cap \mathcal{E}.$$

Thus $\mathcal{B}$ consists of upper triangular affine automorphisms. The Jung–van der Kulk theorem gives the full-group equality

$$G = \mathcal{A} *_{\mathcal{B}} \mathcal{E}. \quad (2.1)$$

We use the reduced-word formulation: a nonempty alternating product of elements of $\mathcal{A} \setminus \mathcal{B}$ and $\mathcal{E} \setminus \mathcal{B}$ cannot belong to $\mathcal{B}$. The exact input is available in [3, Section 2.1] and [1, Section 2, Facts 1–2 and Proposition 1]. Equation (2.1) applies to the entire polynomial automorphism group, so the following argument does not restrict a commuting map to a chosen normal-form family.

2.1 The tree and the axis

Construct a graph $\mathcal{T}$ with vertex set $G/\mathcal{A} \sqcup G/\mathcal{E}$ and edges $G/\mathcal{B}$: the edge $g\mathcal{B}$ joins $g\mathcal{A}$ and $g\mathcal{E}$. The graph is connected because $\mathcal{A}$ and $\mathcal{E}$ generate G . A non-backtracking closed edge path would give a nonempty alternating word of non- $\mathcal{B}$ elements of the two factors lying in $\mathcal{B}$, contrary to the reduced-word theorem. Therefore $\mathcal{T}$ is a tree. Give each edge length one. Left multiplication by G acts by isometries and preserves the two vertex types, so no group element inverts an edge.

Put

$$\tau(x, y) = (y, x), \quad H_d = H_{1,d}, \quad e_d(x, y) = (-x + y^d, y).$$

Then $H_d = \tau e_d$. Define the monic word

$$F_0 = H_7 H_{15} H_{15} H_7 = \tau e_7 \tau e_{15} \tau e_{15} \tau e_7. \quad (2.2)$$

Here F_0 denotes the monic word, not the undefined value $a = 0$ of the family (1.4); it equals F_a when $a = 1$. Each displayed factor of the last expression lies outside $\mathcal{B}$, and the first and last factors belong to different groups. Thus the word is cyclically reduced.

Starting from the base edge $\mathcal{B}$, the successive prefix edges are

$$\mathcal{B}, \quad \tau\mathcal{B}, \quad H_7\mathcal{B}, \quad H_7\tau\mathcal{B}, \quad \dots, \quad F_0\mathcal{B}. \quad (2.3)$$

Consecutive edges meet without backtracking. Repeating this path by F_0^j for $j \in \mathbb{Z}$ introduces no backtracking at its joins because the word is cyclically reduced. The result is a bi-infinite geodesic $\mathcal{L}$, translated by F_0 through eight edge lengths. Write $C_G(F_0) = \{h \in G : hF_0 = F_0h\}$ for its centralizer.

Lemma 2.1. *The line $\mathcal{L}$ is the minimal-displacement set of F_0 . Every member of the centralizer $C_G(F_0)$ preserves $\mathcal{L}$ and acts on it as a translation by an even integer number of edges.*

Proof. For a point z in the metric tree, let q be its projection to $\mathcal{L}$. If z is off the line, its branch and the branch containing F_0z attach at the distinct points q and F_0q . The unique path from z to F_0z therefore has length

$$d(z, F_0z) = 8 + 2d(z, \mathcal{L}). \quad (2.4)$$

The same equality holds for $z \in \mathcal{L}$. Hence $\mathcal{L}$ is precisely the set on which displacement is minimal. If $hF_0 = F_0h$, then $d(hz, F_0hz) = d(z, F_0z)$, so h preserves this line. An orientation-reversing isometry of a line conjugates a nonzero translation to its inverse. Since h commutes with F_0 , its restriction cannot reverse orientation. It is therefore a translation, possibly the identity. It takes vertices to vertices of the same type, which forces an even edge displacement. $\square$

2.2 An intrinsic label on each elementary turn

At an $\mathcal{E}$ -vertex of $\mathcal{L}$ represented by $g\mathcal{E}$, its two incident axis edges have representatives $ge_1\mathcal{B}$ and $ge_2\mathcal{B}$ with $e_1, e_2 \in \mathcal{E}$. Define

$$\text{turn}(g\mathcal{E}) = \deg(e_1^{-1}e_2). \quad (2.5)$$

The degree of a polynomial map means the maximum of the total degrees of its coordinates.

Lemma 2.2. *The label (2.5) is well defined for the unordered pair of incident edges, is at least two, and is preserved by the action of every element of G .*

Proof. The two edges are distinct, so $e_1^{-1}e_2 \notin \mathcal{B}$. It is an elementary map and hence has degree at least two. Changing the vertex representative g to ge replaces each e_i by $e^{-1}e_i$ and leaves their relative product unchanged. Changing the edge representatives on the right by $b_1, b_2 \in \mathcal{B}$ replaces the relative map by $b_1^{-1}e_1^{-1}e_2b_2$. Left or right multiplication of a nonlinear elementary map by a triangular affine map preserves its degree: the substitution in y is invertible affine, its leading term is rescaled by a nonzero scalar, and all other added terms have degree at most one. Exchanging the edges replaces the relative elementary map by its inverse, whose degree is the same. Finally, left multiplication by any element of G leaves the relative pair of representatives unchanged. These checks prove all assertions. $\square$

Along $\mathcal{L}$, the labels in one orientation are

$$\dots, 7, 15, 15, 7, 7, 15, 15, 7, \dots \quad (2.6)$$

This sequence has least positive shift period four: shifts one and two change its first entry, while shift three changes its second.

Proposition 2.3. *For every $h \in C_G(F_0)$ there is $j \in \mathbb{Z}$ such that $h_0 = hF_0^{-j}$ fixes $\mathcal{L}$ pointwise and still centralizes F_0 .*

Proof. By Lemma 2.1, h acts by a translation. It preserves the turn labels by Lemma 2.2. Its displacement must therefore be a multiple of four $\mathcal{E}$ -vertices, or eight edges, by (2.6). The map F_0 realizes exactly this positive displacement. Removing the corresponding signed power F_0^j makes the action on $\mathcal{L}$ trivial, and commutation with F_0 is preserved. $\square$

The proposition controls every polynomial commuting map, including those of arbitrarily large degree. In particular, it rules out a shorter translation generated by a hidden root of the original word.

3 The complete centralizer and geometric reversing coset

It remains to compute the centralizers whose action on $\mathcal{L}$ is trivial. For $\zeta \in \mathbb{C}^\times$ write $S_\zeta(x, y) = (\zeta x, \zeta^{-1}y)$, and let $\mu_8 = \{\zeta \in \mathbb{C} : \zeta^8 = 1\}$.

Proposition 3.1. *The full centralizer of the monic word is*

$$C_G(F_0) = \{S_\zeta F_0^j : \zeta \in \mu_8, j \in \mathbb{Z}\} \cong \mu_8 \times \mathbb{Z}. \quad (3.1)$$

Proof. Take the pointwise-axis centralizer h_0 supplied by Proposition 2.3. The edges $\mathcal{B}$ and $\tau\mathcal{B}$ belong to (2.3), so their stabilizers force

$$h_0 \in \mathcal{B} \cap \tau\mathcal{B}\tau^{-1} = \{(x, y) \mapsto (\alpha x + c, \beta y + e) : \alpha\beta \neq 0\}. \quad (3.2)$$

The equality follows by intersecting upper and lower triangular affine maps. At this point both translations are allowed.

Let $(d_1, d_2, d_3, d_4) = (7, 15, 15, 7)$, put $P_0 = \text{id}$, and set $P_i = H_{d_1} \cdots H_{d_i}$ for $1 \leq i \leq 4$. Both $P_i \mathcal{B}$ and $P_i \tau \mathcal{B}$ lie on the axis for $0 \leq i \leq 4$; the last pair belongs to the next translated segment. Thus each $h_i = P_i^{-1} h_0 P_i$ is diagonal affine, and

$$h_i = H_{d_i}^{-1} h_{i-1} H_{d_i} \quad (1 \leq i \leq 4), \quad h_4 = F_0^{-1} h_0 F_0 = h_0.$$

For $D(x, y) = (\alpha x + c, \beta y + e)$, direct substitution gives

$$H_d^{-1} D H_d(x, y) = ((\alpha y + c)^d - \beta y^d + \beta x - e, \alpha y + c). \quad (3.3)$$

For this map to be diagonal affine, the coefficient of y^d must vanish, giving $\beta = \alpha^d$. The coefficient of y^{d-1} is then $d\alpha^{d-1}c$, so $c = 0$ in characteristic zero. The degrees used here are seven and fifteen. The resulting map is $(\beta x - e, \alpha y)$. Applying the next factor forces its first translation $-e$ to vanish. Hence both translations in (3.2) are zero.

The four successive conjugations now interchange the two diagonal scalars and give precisely

$$\beta = \alpha^7, \quad \alpha = \beta^{15}, \quad \beta = \alpha^{15}, \quad \alpha = \beta^7. \quad (3.4)$$

The first and third equations imply $\alpha^8 = 1$, and then $\beta = \alpha^{-1}$. Conversely, those two conditions satisfy all four equations. Moreover, $H_d^{-1} S_\zeta H_d = S_{\zeta^{-1}}$ for $d \in \{7, 15\}$ and $\zeta^8 = 1$, so every S_ζ really commutes with F_0 . Proposition 2.3 now gives equality in (3.1). The finite diagonal subgroup commutes with $\langle F_0 \rangle$. Their intersection is trivial, since every nonzero power of F_0 has a nonzero axis translation. This proves the direct-product assertion as well. $\square$

Corollary 3.2. *Every polynomial reversor of F_0 over $\mathbb{C}$ belongs to the coset*

$$\text{Rev}_{\mathbb{C}}(F_0) = \{S_\zeta F_0^j \tau : \zeta^8 = 1, j \in \mathbb{Z}\}. \quad (3.5)$$

Proof. The identity $\tau H_d \tau = H_d^{-1}$ holds by direct substitution. Since the ordered word in (2.2) is a palindrome, $\tau F_0 \tau = F_0^{-1}$. For any reversor R , the map $R\tau$ centralizes F_0 . Conversely, multiplying any centralizer by τ gives a reversor. Proposition 3.1 proves the equality. $\square$

No order condition on R was used in this argument. The existence of one involution made the coset accessible; the full centralizer calculation made the answer exhaustive.

4 Scalar twisting and descent over the original field

We now prove Theorem 1.2. Fix a number field L and $a \in L^\times$. Choose an embedding $L \hookrightarrow \mathbb{C}$ and $u \in \mathbb{C}$ with $u^8 = a$, and set $A_u = \text{diag}(u, u^{-1})$. These choices are proof devices for a geometric classification; the statement remains over L .

The identities

$$A_u^{-1} H_d A_u^{-1} = H_{u^{d+1}, d}, \quad A_u H_d A_u = H_{u^{-(d+1)}, d} \quad (4.1)$$

follow by evaluating the two coordinates. Inserting alternating A_u and A_u^{-1} between the four factors gives

$$\begin{aligned} A_u^{-1} F_0 A_u &= (A_u^{-1} H_7 A_u^{-1})(A_u H_{15} A_u)(A_u^{-1} H_{15} A_u^{-1})(A_u H_7 A_u) \\ &= H_{a, 7} H_{a^{-2}, 15} H_{a^2, 15} H_{a^{-1}, 7} = F_a. \end{aligned} \quad (4.2)$$

The conjugated swap is

$$A_u^{-1}\tau A_u = R_{t_0}, \quad t_0 = u^2, \quad t_0^8 = a^2. \quad (4.3)$$

Since A_u commutes with all S_ζ , the preceding full centralizer and reversing-coset equalities conjugate to the same equalities for F_a . In particular,

$$\text{Rev}_{\mathbb{C}}(F_a) = \{F_a^j R_t : j \in \mathbb{Z}, t^8 = a^2\}. \quad (4.4)$$

Indeed, $S_\zeta R_{t_0} = R_{\zeta^{-1}t_0}$, and $\zeta^{-1}t_0$ runs through all eight roots of a^2 . Thus roots with $t^4 = -a$ are retained; the necessary scalar equation is not the stronger equation $t^4 = a$.

Proof of Theorem 1.2. Let $R \in \text{Rev}_L(F_a)$. After the chosen embedding, (4.4) writes $R = F_a^j R_t$ for some $j \in \mathbb{Z}$ and $t \in \mathbb{C}^\times$ with $t^8 = a^2$. For either sign of j , the map F_a^{-j} is a polynomial automorphism over L . Therefore

$$F_a^{-j} R = R_t \in \text{Aut}_L(\mathbb{A}^2).$$

The coefficient of x in the second coordinate of R_t is t , so $t \in L^\times$. This proves necessity in every integer component, regardless of the degree of the original R . Conversely, a root $t \in L^\times$ gives R_t by (4.4), and each $F_a^j R_t$ is then defined over L and reverses F_a . This proves the full equality (1.5) and its existence criterion. $\square$

Lemma 4.1. *Let E be any characteristic-zero field, let $a, t \in E^\times$, and suppose $t^8 = a^2$. Then the affine involution R_t reverses F_a over E .*

Proof. In an algebraic closure of E , choose $u^8 = a$ and put $t_0 = u^2$. The coordinate identities (4.1)–(4.3) hold over this closure. Put $\zeta = t_0/t$; then $\zeta^8 = 1$. The identity $H_d^{-1} S_\zeta H_d = S_{\zeta^{-1}}$ for $d = 7, 15$ proves that S_ζ commutes with F_0 . It also commutes with A_u , and hence with $F_a = A_u^{-1} F_0 A_u$. Thus $S_\zeta R_{t_0} = R_t$ reverses F_a . Both sides of the reversal identity have coefficients in E , so the identity already holds over E . Finally, $R_t^2 = \text{id}$ by its two-coordinate formula. $\square$

The lemma constructs the witnesses over completions without requiring an embedding of a completion into $\mathbb{C}$. Only the global classification uses such an embedding for a number field. Removing the original-field native power in the proof of Theorem 1.2 is also essential: coefficient cancellation in a high-degree expression $F_a^j R_t$ cannot evade that operation.

5 The Wang class at every original completion

We give the full arithmetic check, including the dyadic and infinite places. The exceptional class itself is classical; the precise field, exponent, and class appear in the remark following Proposition 2.1 of the accessible preprint of [4].

Proposition 5.1. *For $K = \mathbb{Q}(\sqrt{7})$ one has*

$$16 \in K_v^{\times 8} \quad \text{for every } v \in \Omega_K, \quad 16 \notin K^{\times 8}. \quad (5.1)$$

Proof. First let p be an odd rational prime. At least one of $2, -2, -1$ is a square modulo p . To see this, write χ for the quadratic character of $\mathbb{F}_p^\times$. If neither 2 nor -1 is a square, then $\chi(-2) = \chi(-1)\chi(2) = 1$. These three elements are nonzero modulo p , so each square root found there lifts to $\mathbb{Q}_p$ by Hensel's lemma: the derivative of $X^2 - b$ is nonzero at a nonzero root modulo the odd prime p . If $r^2 = 2$ or $r^2 = -2$, then $r^8 = 16$. If $i^2 = -1$, then

$$(1+i)^2 = 2i, \quad (1+i)^4 = -4, \quad (1+i)^8 = 16.$$

Thus $\mathbb{Q}_p$ contains an eighth root of 16 for every odd p . Each completion K_v above p contains $\mathbb{Q}_p$, proving the required statement at all these places, including those above the ramified prime seven.

Next consider the rational prime two. The element 7 is not a square in $\mathbb{Q}_2$: an odd square is congruent to 1 modulo 8. Hence

$$K \otimes_{\mathbb{Q}} \mathbb{Q}_2 = \mathbb{Q}_2(\sqrt{7})$$

is a field, and K has exactly one place above two. On the other hand -7 is a square in $\mathbb{Q}_2$. The strong Hensel criterion for $f(X) = X^2 + 7$ at $X = 1$ applies because

$$v_2(f(1)) = 3 > 2v_2(f'(1)) = 2,$$

where $v_2(2) = 1$. Choose $b \in \mathbb{Q}_2$ with $b^2 = -7$. In the original dyadic completion, $i = \sqrt{7}/b$ satisfies $i^2 = -1$. Consequently $1 + i$ is an eighth root of 16 there. Equivalently, that completion is $\mathbb{Q}_2(i)$.

The field K is real quadratic. Both infinite completions are $\mathbb{R}$, and each contains $t = \sqrt{2}$ with $t^8 = 16$. There are no complex places.

Finally, any real solution of $t^8 = 16$ is $\sqrt{2}$ or $-\sqrt{2}$. Neither lies in $\mathbb{Q}(\sqrt{7})$. Indeed, if $(b + c\sqrt{7})^2 = 2$ with $b, c \in \mathbb{Q}$, comparison of the coefficient of $\sqrt{7}$ forces $bc = 0$. The two alternatives would require a rational square to equal 2 or $2/7$, respectively. Both numbers have odd 2-adic valuation, which is impossible for a rational square. This proves (5.1). $\square$

Table 1: The local roots used for $K = \mathbb{Q}(\sqrt{7})$. Every row concerns the original completions. The complete proofs are in Proposition 5.1; K has no complex places.

Places	An eighth root of 16
All v above odd p	A square root of 2 or -2 in $\mathbb{Q}_p$, or $1 + i$ when $i^2 = -1$ in $\mathbb{Q}_p$; at least one option exists.
The unique v above 2	$1 + i$ in $K_v = \mathbb{Q}_2(\sqrt{7}) = \mathbb{Q}_2(i)$.
The two real places	$\sqrt{2}$ in each completion $\mathbb{R}$.

Proof of Theorem 1.1. Take $a = 4$ in (1.4); the resulting F_a is the word (1.3), and $a^2 = 16$. At each original place v , Proposition 5.1 supplies $t_v \in K_v^\times$ with $t_v^8 = 16$. Lemma 4.1 gives

$$R_v(x, y) = (t_v^{-1}y, t_v x), \quad R_v^2 = \text{id}, \quad R_v F R_v^{-1} = F^{-1}$$

over that same K_v . If any polynomial reversor existed over K , Theorem 1.2 would force an eighth root of 16 in K , contrary to Proposition 5.1. This excludes all global polynomial reversors, not only the local affine formulas. $\square$

6 A two-factor control and the scope of the result

The two-factor analogue shows why a failed affine equation is not a global obstruction. Consider

$$T = H_{4,7} H_{1/4,7}, \quad X = y^7/4 - x.$$

Then $T(x, y) = (X, 4X^7 - y)$. Define polynomial involutions over K by

$$E_1(x, y) = (y^7/4 - x, y), \quad E_2(x, y) = (x, 4x^7 - y).$$

Direct substitution gives $E_1^2 = E_2^2 = \text{id}$ and $T = E_2E_1$. Consequently

$$E_1TE_1 = E_1E_2 = T^{-1}.$$

Thus T is globally reversible by a nonlinear polynomial involution. In contrast, the second coordinate of R_tTR_t is $(t^8/4)x^7 - y$, whereas the second coordinate of T^{-1} is $4x^7 - y$. A reversor of the selected form R_t would therefore require $t^8 = 16$, which has no solution in K by Proposition 5.1. Its repeated degree cycle does not give the primitive four-factor translation restriction established for F_0 . This control identifies the role of the whole-group argument: one must exclude the other polynomial reversing components, not only a chosen scalar formula.

Theorem 1.1 gives a negative answer to (1.1) for one original nonlinear map. The supporting family criterion is part of that same construction. Neither theorem classifies all reversible Hénon words or asserts that this example has minimal degree or shortest possible word length. No integrality or good-reduction hypothesis is imposed. The arithmetic obstruction, the amalgam methods, and scalar-twist descent remain classical inputs; the conclusion concerns their realization for the intrinsic symmetry relating F to F^{-1} .

Preparation and verification

This manuscript was prepared with AI assistance and current-team internal mathematical and source review. Those checks are not external peer review or publication acceptance. The complete construction and exclusion arguments are included above; their classical inputs are the cited amalgam theorem and basic local-field facts, including Hensel's lemma. No numerical experiment or computer-algebra verification is used as a premise.

References

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